



An example showcasing the features of the msheet package

1 The msheet class (msheet.cls)

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2 The msheet style (msheet.sty)

A few tasks originally placed by [Gilbert Strang](#) are chosen to illustrate the package features.

Aufgabe 1 *A subsection like task environment.*

Compute the average velocity between $t = 5$ and $t = 8$.

- | | | |
|---------------------|--------------------|-----------------------------|
| 1. $f(t) = 6t$ | 2. $f(t) = 6t + 2$ | 3. $f(t) = \frac{1}{2}at^2$ |
| 4. $f(t) = t - t^2$ | 5. $f(t) = 6$ | |

Aufgabe *And the same as starred version.*

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| d) $f(t) = t - t^2$ | e) $f(t) = 6$ | |

Aufgabe 2 *A paragraph like task environment.*

- | | | |
|---------------------|--------------------|-----------------------------|
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| d) $f(t) = t - t^2$ | e) $f(t) = 6$ | |

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»» Bitte wenden! »»

1. When you jump and fall back your height is $y = 2t - t^2$ in the right units.
 - 1.1. Graph this parabola and its slope.
 - 1.2. Find the time in the air and the maximum height.
 - 1.3. Prove: Half of the time you are above $y = \frac{3}{4}$. tip 1 solution 1

Basketball players hang in the air partly because of 1.3..
2. If the ball on the unit circle reaches t degrees at the time t , find its position and speed and upward velocity. recommended

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Basketball players hang in the air partly because of c).

Aufgabe 2 If the ball on the unit circle reaches t degrees at the time t , find its position and speed and upward velocity. recommended

Theorem 1 – l'Hôpitals rule

Suppose $f(x)$ and $g(x)$ both approach zero as $x \rightarrow a$. Then $\frac{f(x)}{g(x)}$ approaches the same limit as $\frac{f'(x)}{g'(x)}$, if that second limit exists:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.$$

Experiment – The gravitational force

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Example

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A golden titled box

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